



ព្រះរាជាណាចក្រកម្ពុជា
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Course: Mobile App Dev

Lab01 - Intro to Flutter

Name of Students

ID of Students

Score

MA SOVITCHEA

e20221575

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Group: I4-GIC-C

Lecturer: Mr. HOK Tin

Academic year: 2025 ~ 2026

Lab01's Solution

Exercise 1 — Environment Validation

- **Command run:**

flutter doctor

- **Initial output issues found:**

```
E:\I4-S2\Mobile App\Lab1>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[✓] Flutter (Channel stable, 3.41.4, on Microsoft Windows [Version 10.0.22631.6199], locale en-US)
[✓] Windows Version (11 Home Single Language 64-bit, 23H2, 2009)
[X] Android toolchain - develop for Android devices
    X Unable to locate Android SDK.
      Install Android Studio from: https://developer.android.com/studio/index.html
      On first launch it will assist you in installing the Android SDK components.
      (or visit https://flutter.dev/to/windows-android-setup for detailed instructions).
      If the Android SDK has been installed to a custom location, please use
      `flutter config --android-sdk` to update to that location.

[✓] Chrome - develop for the web
[X] Visual Studio - develop Windows apps
    X Visual Studio not installed; this is necessary to develop Windows apps.
      Download at https://visualstudio.microsoft.com/downloads/.
      Please install the "Desktop development with C++" workload, including all of its
      default components
[✓] Connected device (3 available)
[✓] Network resources

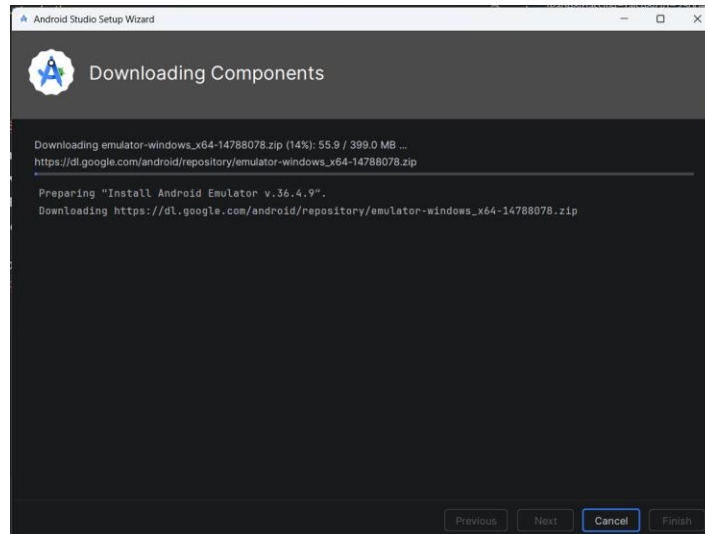
! Doctor found issues in 2 categories.

E:\I4-S2\Mobile App\Lab1>
```

- **Issue 1: Android toolchain**
- **Problem: Flutter could not locate the Android SDK.**

Steps to resolve:

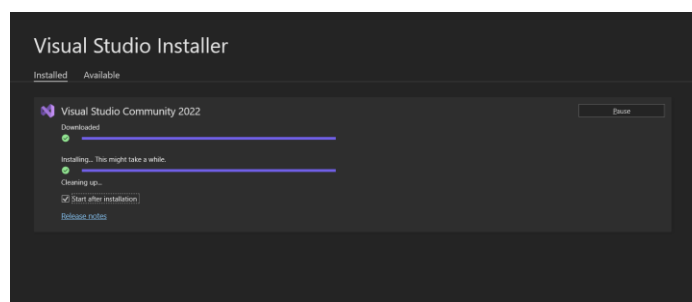
1. Visited developer.android.com/studio
2. Downloaded android-studio-panda2-windows.exe (Windows 64-bit, Recommended)
3. Ran the installer with default settings, ensuring Android Virtual Device was checked
4. Completed the Android Studio setup wizard using Standard install type
5. Let Android Studio download and install the Android SDK components
6. Ran flutter doctor android-licenses and accepted all licenses by typing y
7. Re-ran flutter doctor – Android toolchain now shows 



- **Issue 2: Visual Studio**
- **Problem: Visual Studio not installed, which is required for Windows desktop app development.**

Steps to resolve:

1. Visited visualstudio.microsoft.com/downloads
2. Downloaded Visual Studio Community (free)
3. Ran the installer
4. Selected the “Desktop development with C++” workload
5. Clicked Install and waited for the download to complete (~7-8 GB)
6. Re-ran flutter doctor – Visual Studio now shows 



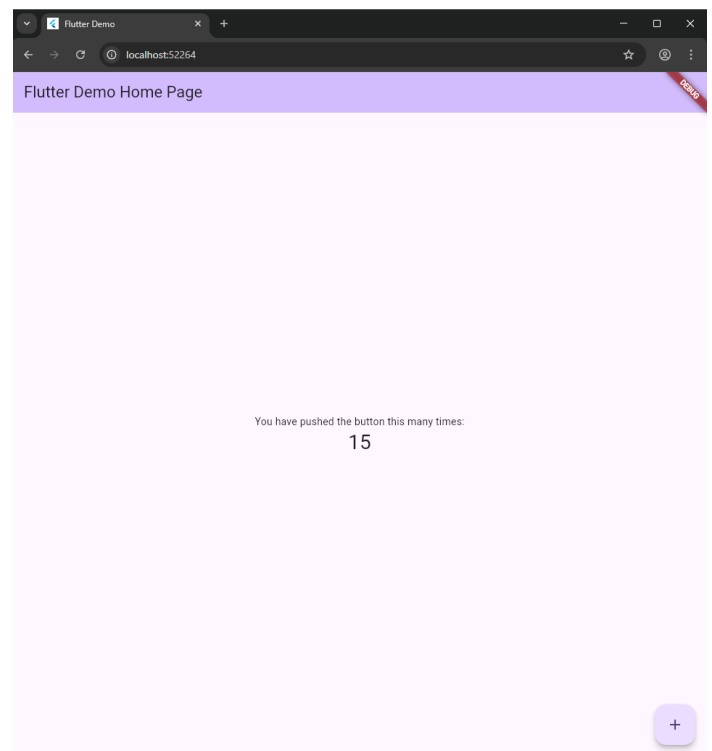
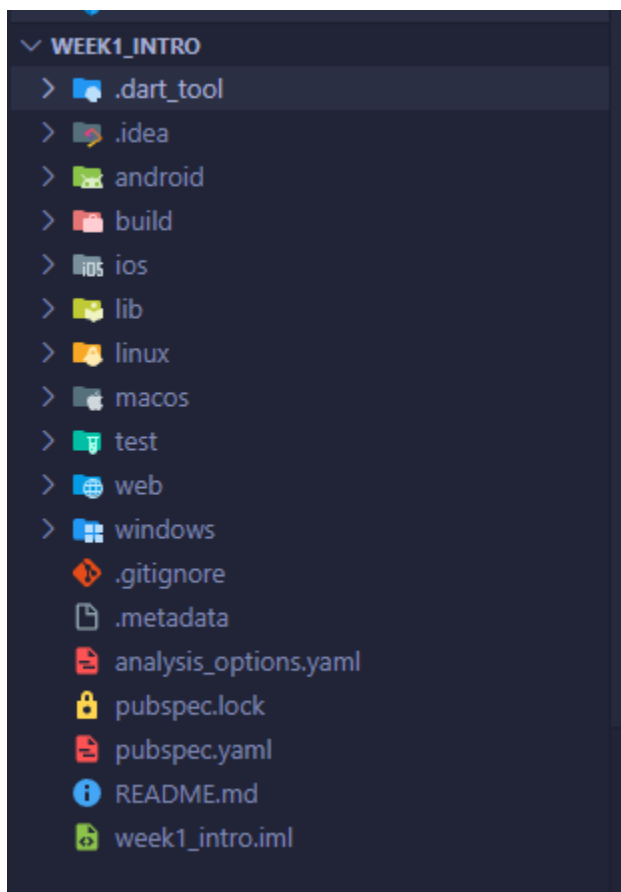
- Resolved issues Output:

```
E:\I4-S2\Mobile App\Lab1>flutter doctor
Doctor summary (to see all details, run flutter doctor -v):
[v] Flutter (Channel stable, 3.41.4, on Microsoft Windows [Version 10.0.22631.6199], locale en-US)
[v] Windows Version (11 Home Single Language 64-bit, 23H2, 2009)
[v] Android toolchain - develop for Android devices (Android SDK version 36.1.0)
[v] Chrome - develop for the web
[v] Visual Studio - develop Windows apps (Visual Studio Community 2022 17.14.27 (February 2026))
[v] Connected device (3 available)
[v] Network resources

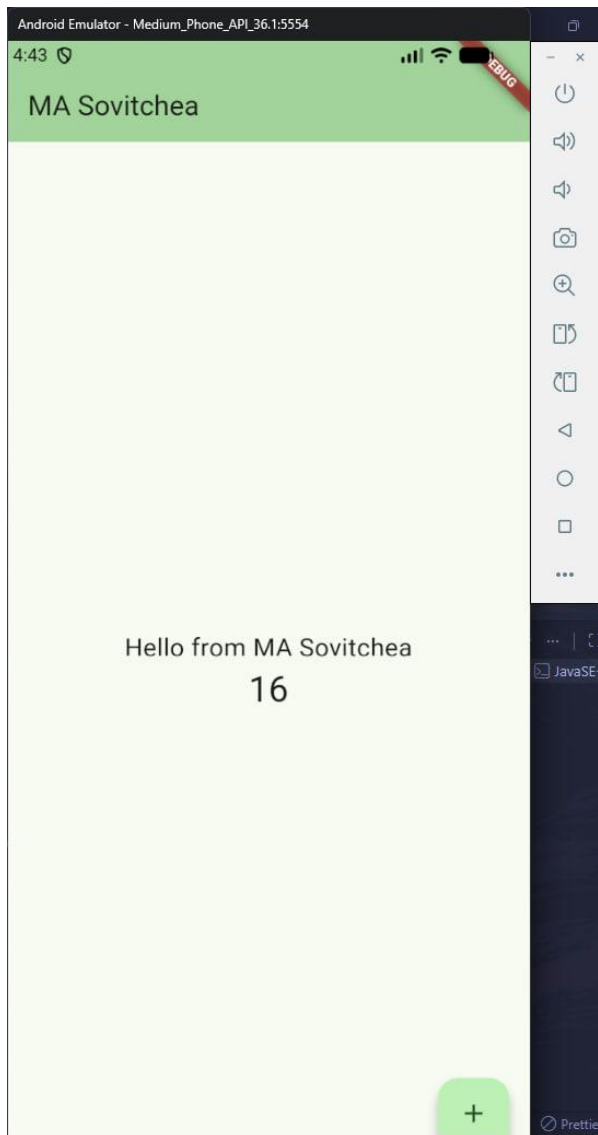
• No issues found!

E:\I4-S2\Mobile App\Lab1>
```

Exercise 2 — First Project



Exercise 3 — Customize the Home Screen



Exercise 4 — Explore pubspec.yaml

```
24 # Dependencies specify other packages that your package needs in order to work.
25 # To automatically upgrade your package dependencies to the latest versions
26 # consider running `flutter pub upgrade --major-versions`. Alternatively,
27 # dependencies can be manually updated by changing the version numbers below to
28 # the latest version available on pub.dev. To see which dependencies have newer
29 # versions available, run `flutter pub outdated`.
30 dependencies:
31   flutter:
32     sdk: flutter
33   http: ^1.2.0
34
35 # The following adds the Cupertino Icons font to your application.
36 # Use with the CupertinoIcons class for iOS style icons.
37 cupertino_icons: ^1.0.8
```

```
E:\I4-S2\Mobile App\Lab1\week1_intro>flutter pub get
Resolving dependencies...
Downloading packages...
  meta 1.17.0 (1.18.1 available)
Got dependencies!
1 package has newer versions incompatible with dependency constraints.
Try `flutter pub outdated` for more information.

E:\I4-S2\Mobile App\Lab1\week1_intro>
```

Exercise 5 — Widget Exploration

